Ian Waudby-Smith

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Academic Positions

University of California, Berkeley — **Berkeley, CA**
Postdoctoral Miller Fellow — *2024–2027*
Host: Michael I. Jordan

Education

Carnegie Mellon University — **Pittsburgh, PA**
PhD, Statistics — *2024*
Thesis: *Advances in anytime-valid sequential inference*
Advisor: Aaditya Ramdas
Recipient of the Umesh K. Gavaskar Memorial Dissertation Award

University of Waterloo — **Waterloo, Ontario**
BMath, Pure Mathematics & Statistics — *2018*
Dean's Honours List

Papers

Publications

[1] ($\alpha\beta$) Johannes Ruf and **Ian Waudby-Smith**. Concentration inequalities for strong laws and laws of the iterated logarithm. *The Bulletin of the London Mathematical Society (to appear)*, 2026.

[2] **Ian Waudby-Smith**, Martin Larsson, and Aaditya Ramdas. Distribution-uniform strong laws of large numbers. *The Annals of Applied Probability (to appear)*, 2026.

[3] **Ian Waudby-Smith**, David Arbour, Ritwik Sinha, Edward H. Kennedy, and Aaditya Ramdas. Time-uniform central limit theory and asymptotic confidence sequences. *The Annals of Statistics*, 52(6):2613–2640, 2024.

[4] **Ian Waudby-Smith**, Lili Wu, Aaditya Ramdas, Nikos Karampatziakis, and Paul Mineiro. Anytime-valid off-policy inference for contextual bandits. *ACM/IMS Journal of Data Science* ***(Invited contribution)***, 1(3):1–42, 2024.

[5] **Ian Waudby-Smith** and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology* ***(Discussion paper)***, 86(1):1–27, 2024.

[6] **Ian Waudby-Smith**, Zhiwei Steven Wu, and Aaditya Ramdas. Extensions of randomized response for private confidence sets. *International Conference on Machine Learning* ***(Oral presentation)***,

2023.

[7] Akash V. Maharaj, Ritwik Sinha, David Arbour, **Ian Waudby-Smith**, Simon Z. Liu, Moumita Sinha, Raghavendra Addanki, Aaditya Ramdas, Manas Garg, and Viswanathan Swaminathan. Anytime-valid confidence sequences in an enterprise A/B testing platform. *The ACM World Wide Web Conference*, 2024.

[8] **Ian Waudby-Smith**, Philip B. Stark, and Aaditya Ramdas. RiLACS: Risk limiting audits via confidence sequences. In *International Joint Conference on Electronic Voting* ***(Best paper award)***, pages 124–139. Springer, 2021.

[9] **Ian Waudby-Smith** and Aaditya Ramdas. Confidence sequences for sampling without replacement. *Advances in Neural Information Processing Systems* ***(Spotlight paper)***, 33:20204–20214, 2020.

[10] **Ian Waudby-Smith**, A Simon Pickard, Feng Xie, and Eleanor M. Pullenayegum. Using both time tradeoff and discrete choice experiments in valuing the EQ-5D: Impact of model misspecification on value sets. *Medical Decision Making*, 2020.

[11] **Ian Waudby-Smith**, Nam Tran, Joel A. Dubin, and Joon Lee. Sentiment in nursing notes as an indicator of out-of-hospital mortality in intensive care patients. *PloS one*, 13(6), 2018.

Preprints

[12] Chase Mathis and **Ian Waudby-Smith**. Confidence horizons. *Under review at the Journal of the Royal Statistical Society, Series B.*

[13] Keegan Harris, Brian W. Lee, **Ian Waudby-Smith**, Philip Amortila, Nika Haghtalab, and Michael I. Jordan. Post-training at the edge of detectability: A game-theoretic approach to fine-tuning. *Under review at the International Conference on Learning Representations (ICLR)*, 2026.

[14] ($\alpha\beta$) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith**. Gaffke's confidence interval for the mean of bounded data is inadmissible but asymptotically efficient. *arXiv preprint arXiv:2607.18661.*

[15] ($\alpha\beta$) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith**. Combining e-values via demi-supermartingales. *Under review at the Annals of Applied Probability.*

[16] Ricardo J. Sandoval*, **Ian Waudby-Smith***, and Michael I. Jordan. Multi-armed sequential hypothesis testing by betting. *arXiv preprint arXiv:2603.17925.*

[17] Ben Chugg*, Etienne Gauthier*, Michael I. Jordan, Aaditya Ramdas, and **Ian Waudby-Smith**. Post-hoc large sample statistical inference. *Under review at the Journal of the Royal Statistical Society, Series B*, 2026.

[18] Ricardo J. Sandoval, Sivaraman Balakrishnan, Avi Feller, Michael I. Jordan, and **Ian Waudby-Smith**. On nonasymptotic confidence intervals for treatment effects in randomized experiments. *Major revision at Biometrika.*

[19] **Ian Waudby-Smith**, Ricardo J. Sandoval, and Michael I. Jordan. Instance-log-optimality of portfolio-based e-processes and their sequential hypothesis tests. *R&R at the Annals of Statistics*, 2025.

[20] **Ian Waudby-Smith**, Martin Larsson, and Aaditya Ramdas. Nonasymptotic and distribution-uniform Komlós-Major-Tusnády approximation. *Major revision at Probability Theory and Related Fields.*

[21] **Ian Waudby-Smith**, Edward H. Kennedy, and Aaditya Ramdas. Distribution-uniform large sample anytime-valid inference and the Robbins–Siegmund distributions. *Major revision at Biometrika.*

($\alpha\beta$) Alphabetical ordering.
*Co-lead author.

Teaching Experience

Carnegie Mellon University **Pittsburgh, PA**
Graduate Teaching Assistant *2019–22*

- 36-708: Statistical Methods in Machine Learning (x2)
- 36-462: Data Mining
- 36-401: Modern Regression
- 36-731: Foundations of Causal Inference
- 36-732: Modern Causal Inference
- 10-880: Game-theoretic Probability, Statistics, and Learning

Other Experience

Google Research **New York, NY**
Student Researcher *Jun–Aug 2023*
Mentors: Jean Pouget-Abadie & Jennifer Brennan

Microsoft Research **Redmond, WA**
Research Intern *May–Aug 2022*
Mentor: Paul Mineiro

Adobe Research **San Jose, CA**
Research Intern *Jun–Aug 2020*
Mentors: David Arbour & Ritwik Sinha

The Hospital for Sick Children (SickKids) **Toronto, Ontario**
Research Student *Apr–Aug 2019*
Mentor: Eleanor Pullenayegum

Health Data Science Lab, University of Waterloo **Waterloo, Ontario**
Research Assistant *2016–18*
Mentors: Joel Dubin & Joon Lee

Department of Statistics, University of Waterloo **Waterloo, Ontario**
Research Assistant *Apr–Aug 2017*
Mentor: Pengfei Li

Department of Pure Mathematics, University of Waterloo **Waterloo, Ontario**
Research Assistant *Jan - Apr 2017*
Mentor: Yu-Ru Liu

Cancer Care Ontario **Toronto, Ontario**
Student Analyst *Jan–Apr 2016*
Mentor: Zhihui (Amy) Liu

Professional Service

Thesis committee member

- Tyron Lardy (PhD in Statistics, Leiden University)
- Noah Liniger (MSc in Data Science, ETH Zurich)

Reviewer

- Annals of Statistics
- Bernoulli
- Biometrika
- Electronic Journal of Probability
- Electronic Journal of Statistics
- Journal of the American Statistical Association
- Journal of the Royal Statistical Society, Series B
- Journal of the Royal Statistical Society, Series C
- Management Science
- New England Journal of Data Science
- Sankhya A.
- Symposium on Theory of Computing (STOC)

University of California, Berkeley **Berkeley, CA**
Volunteer

- Miller symposium organizing committee
- Incoming Miller fellow initiative

Carnegie Mellon University **Pittsburgh, PA**
Volunteer

- Organizer of the Statistical Machine Learning Reading Group (SMLRG)
- Women in Data Science (WiDS) conference volunteer
- Computing committee student representative
- Incoming PhD student mentor

Awards

Institute of Mathematical Statistics **Berkeley, CA**
New Researcher Travel Award *2026*

Miller Institute for Basic Research in Science **Berkeley, CA**
Miller fellowship *2024–'27*

Carnegie Mellon University Department of Statistics and Data Science **Pittsburgh, PA**
2024
Umesh K. Gavaskar Best Dissertation Award

Statistical Society of Canada **St. John's, NL**
Probability Section Student Research Presentation Award *2024*

Amazon Science **Pittsburgh, PA**
Graduate Research Fellowship *2023*

University of Waterloo **Waterloo, Ontario**
Waterloo Statistics Student Conference Presentation Award *2022*

Carnegie Mellon University Department of Statistics and Data Science **Pittsburgh, PA**
Teaching Assistant of the Year *2021*

Adobe Research **Pittsburgh, PA**
PhD Research Gift *2020*

University of Waterloo **Waterloo, Ontario**
David Johnston International Experience Award *2018*

The Natural Sciences and Engineering Research Council of Canada **Waterloo, Ontario**
NSERC Undergraduate Student Research Award *2017*

University of Waterloo **Waterloo, Ontario**
President's Research Award *2016–17*

University of Waterloo **Waterloo, Ontario**
President's Scholarship of Distinction *2014*

Presentations

Short courses

MBZUAI **Abu Dhabi, UAE**
A brief introduction to game-theoretic, safe, anytime-valid inference *2025*
Mini course consisting of 4 lectures

ERC OCEAN retreat **Venice, Italy**
A brief introduction to game-theoretic, safe, anytime-valid inference *2024*
Mini course consisting of 3 lectures

Invited department seminars

Department of Statistics at the London School of Economics **London, UK**
Log-optimality and multi-armed sequential hypothesis testing *2026*

Department of Statistics and Actuarial Science at the University of Waterloo **Waterloo, Ontario**
Log-optimality and multi-armed sequential hypothesis testing *2026*

Department of Statistics at University of California, Davis **Davis, CA**
Log-optimality and multi-armed sequential hypothesis testing *2026*

Department of Statistics at Iowa State University **Ames, IA**
Anytime-valid inference and uniform central limit theory *2026*

Division of Biostatistics at the University of California, Berkeley **Berkeley, CA**
Anytime-valid off-policy inference for contextual bandits *2025*

Department of Statistics at Stanford University **Stanford, CA**
Anytime-valid inference and uniform central limit theory *2025*

Statistics Seminar at the University of Copenhagen **Copenhagen, Denmark**
Anytime-valid off-policy inference for contextual bandits *2023*

Invited talks at conferences, workshops, and research groups

Institute for Operations Research and the Management Sciences (INFORMS) **San Francisco, CA**
Multi-armed log-optimality and sequential testing *2026*

Whendee L. Silver lab at the University of California, Berkeley **Berkeley, CA**
Anytime-valid inference for the environmental sciences *2026*

Inria/Sierra Seminar **Paris, France**
Combining e-values via demi-supermartingales *2026*

Cornell University Rising Stars in Statistics and Data Science Workshop **Ithaca, NY**
Advances in asymptotic anytime-valid inference *2026*

Joint Statistical Meetings (JSM) **Boston, MA**
Asymptotic anytime-valid inference *2026*

Adaptive Experimentation Workshop at University College London (Gatsby) **London, UK**
Log-optimality and multi-armed sequential hypothesis testing *2026*

International Workshop on Sequential Methodologies **Washington, D.C.**
Asymptotic anytime-valid statistical inference *2026*
Recipient of the IMS New Researcher Travel Award

Stanford Causal Inference Reading Group (host: Vasilis Syrgkanis) **Stanford, CA**
On nonasymptotic confidence intervals for treatment effects in randomized experiments *2026*

National University of Singapore Young Mathematical Scientists Forum **Singapore**
Log-optimality of e-processes and sequential tests *2025*

Stanford Data Driven Seminar (host: Stefan Wager) **Stanford, CA**
Log-optimality of e-processes and sequential tests *2025*

MBZUAI-Berkeley Workshop **Abu Dhabi, UAE**
Anytime-valid off-policy inference for contextual bandits *2025*

Inria/Sierra Seminar **Paris, France**
Anytime-valid inference and uniform central limit theory *2025*

International Seminar on Selective Inference **Virtual**
$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X *2024*

CLIMB Workshop **Berkeley, CA**
Election audits via anytime-valid inference *2024*

Workshop on Game-Theoretic Statistical Inference — **Oberwolfach, Germany**
$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X — *2024*

Fienberg Student Research Workshop at Carnegie Mellon University — **Pittsburgh, PA**
Election audits via anytime-valid inference — *2024*

Joint Statistical Meetings — **Toronto, Ontario**
Anytime-valid off-policy inference for contextual bandits — *2023*

International Conference on Machine Learning — **Honolulu, HI**
Extensions of randomized response for private confidence sets — *2023*

Centrum Wiskunde & Informatica (hosts: Peter Grünwald & Wouter Koolen) — **Amsterdam, Netherlands**
Anytime-valid off-policy inference for contextual bandits — *2023*

University of Toronto / The Vector Institute (host: Dan Roy) — **Toronto, Ontario**
Anytime-valid off-policy inference for contextual bandits — *2023*

Copenhagen Causality Lab, University of Copenhagen — **Virtual**
Asymptotic confidence sequences for anytime-valid causal inference — *2023*

Conference on Digital Experimentation (CODE@MIT) — **Cambridge, MA**
Asymptotic confidence sequences for anytime-valid causal inference — *2022*

Microsoft Research Reinforcement Learning Discussion Group — **Virtual**
Anytime-valid contextual bandit inference — *2022*

Computing and Mathematical Sciences, California Institute of Technology (host: Houman Owhadi) — **Virtual**
A brief introduction to safe, anytime-valid inference (SAVI) — *2022*

Microsoft Research — **Virtual**
A brief introduction to safe, anytime-valid inference (SAVI) — *2022*

Safe, Anytime-Valid Inference (SAVI) Workshop — **Eindhoven, Netherlands**
Time-uniform central limit theory and anytime-valid causal inference — *2022*

Carnegie Mellon University Computer Science Theory Lunch — **Pittsburgh, PA**
Estimating means of bounded random variables by betting — *2021*

International Seminar on Distribution-Free Statistics — **Virtual**
Estimating means of bounded random variables by betting — *2021*

E-Vote-ID: The International Conference for Electronic Voting — **Virtual**
RiLACS: Risk-limiting audits via confidence sequences — *2021*

Spotify Experimentation Platform Team — **Virtual**
Doubly robust confidence sequences for sequential causal inference — *2021*

Vinted Science and Analytics Meetup — **Virtual**
Doubly robust confidence sequences for sequential causal inference — *2021*

Contributed conference talks

International Conference on Statistics and Data Science (ICSDS) — **Seville, Spain**, *2025*
Concentration inequalities for strong laws of large numbers

Statistical Society of Canada meeting — **St. John's, NL**, *2024*
Distribution-uniform strong laws of large numbers
Recipient of the Probability Section's Student Presentation Award

International Conference on Statistics and Data Science (ICSDS) — **Lisbon, Portugal**, *2023*
Distribution-uniform anytime-valid inference

Waterloo Student Conference in Statistics, Actuarial Science, and Finance — **Waterloo, Ontario**, *2022*
Estimating means of bounded random variables by betting
Recipient of the Student Presentation Award

TPDP: Theory and Practice of Differential Privacy Workshop — **Baltimore, MD**, *2022*
Locally private nonparametric confidence intervals and sequences

Statistical Society of Canada (SSC) Annual Meeting — **Virtual**, *2022*
Time-uniform central limit theory and anytime-valid causal inference

NeurIPS Workshop on Causal Inference Challenges in Sequential Decision Making — **Virtual**, *2021*
Time-uniform central limit theory and anytime-valid causal inference

Joint Statistical Meetings (JSM) — **Virtual**, *2021*
Doubly robust confidence sequences for sequential causal inference

Joint Statistical Meetings (JSM) — **Virtual**, *2020*
Confidence sequences for sampling without replacement

Statistical Society of Canada (SSC) Annual Meeting — **St. Catherines, Ontario**, *2016*
Multi-state models for chronic kidney disease prevalence projections in Ontario